

CONTACT INFORMATION	Chicago, Illinois, USA  Portfolio,  Scholar  Primary : rajeshr12@zohomail.com , University : rrame12@uic.edu  https://www.linkedin.com/in/rajeshrrajeshr/  https://github.com/its-rajesh
RESEARCH INTERESTS	Broadly interested in audio signal processing and machine learning, with a research emphasis on source separation, interference reduction, distance-aware acoustic modeling, and distributed acoustic sensor networks. More generally, I am interested in problems that involve fundamental signal processing, mathematical modeling, and neural network-based learning.
EDUCATION	University of Illinois Chicago Aug 2024 - Current PhD in Audio Signal Processing at Listening Technology Lab — Area : Audio Signal Processing — CGPA : 4/4 (Current) — Advisor : Prof. Ryan M. Corey  — Selected Courses : <i>Digital Signal Processing II, Adaptive Digital Filters, Statistical Digital Signal Processing, Audio Signal Processing, Information and Learning</i> Indian Institute of Technology Mandi (IIT Mandi) Aug 2021 - April 2024 MS by Research at the School of Computing and Electrical Engineering — Area : Audio Signal Processing - Music Source Separation and Music Bleeding Removal — CGPA : 8.73/10 — Dissertation : <i>Music Source Separation</i> — Advisor : Prof. Padmanabhan Rajan  — Selected Courses : <i>Deep Learning and Applications, Matrix Theory, Applied Optimization, Probability and Random Processes, Pattern Recognition, Programming Practicum for Signal Processing, Digital Image Processing</i> Anna University Aug 2014 - May 2018 Bachelors in Engineering - Electronics and Communication Engineering (ECE) — University Rank (32nd out of 17,534), 2nd in the institute — Highest Honours (top 0.18%), CGPA : 8.9/10 — Top Courses : <i>Digital Signal Processing, Signals and Systems, Linear Algebra, Transforms and Differential Equations, Multivariate Calculus and Complex Integration, Probability and Random Processes</i>
RESEARCH EXPERIENCE	Distance-Aware Source Separation Aug 2025 – Present Conventional source separation systems perform well in constrained domains (e.g., speech or music), but struggle to generalize across acoustic environments where <i>source-to-microphone distance</i> is a dominant factor. This project develops a universal multichannel separation framework that conditions neural models on distance information, enabling reliable discrimination and reconstruction of near and far sources across speech, music, and environmental audio. ★ Neural conditioning via learned distance embeddings and robust spatial features, including ILD, IPD, ITD, multi-resolution energy envelopes, and DRR-based reverberation cues. ★ Project page :  Acoustic Sensor Networks Aug 2025 – Present Investigates distributed acoustic sensing for scalable and energy-efficient audio processing, including source localization, beamforming, interference reduction, and distance-aware separation. The goal is to design coordination algorithms for multi-node microphone arrays that exchange minimal information while achieving performance comparable to centralized systems, with applications in wearables, smart spaces, AR/VR, and IoT audio platforms. ★ Focus on DANSE-style distributed beamforming and neural message-passing architectures (“Neural DANSE”) under synchronization and bandwidth constraints.

- Interference Reduction** Aug 2024 – July 2025
 Developed interference reduction methods that are agnostic to source type and robust to highly reverberant conditions, with particular emphasis on live multitrack recordings where interference exhibits complex, nonlinear, and content-dependent characteristics.
 - ★ Generative and analytical approaches to interference modeling, quantification, and suppression across waveform, STFT, and wavelet representations.
 - ★ Project page : 
- Music Source Separation** Dec 2021 – Apr 2024
 Investigated music source separation beyond the standard 4–5 stem setting, with emphasis on Indian classical music genres such as Carnatic and Hindustani, where existing datasets and models are limited. Explored scalable architectures capable of handling a larger number of interacting sources.
 - ★ Wave-U-Net architectures augmented with graph attention mechanisms.
 - ★ Project page : 
- Interference Reduction in Music Signals** Dec 2022 – Aug 2023
 Addressed microphone bleed in live recordings of Indian Carnatic concerts using a combination of signal processing and learning-based techniques, with emphasis on minimizing correlated interference while preserving perceptual quality for professional mixing and archival applications.
 - ★ Learning-free optimization methods, convolutional autoencoders, Wave-U-Net variants, and graph attention networks.
 - ★ Project page : 

SELECTED PUBLICATIONS

- [C1] **Rajesh R** and Padmanabhan Rajan, “Neural Networks for Interference Reduction in Multi-Track Recordings,” 2023 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), New Paltz, NY, USA, 2023, pp. 1-5.  

IN REVIEW / PRE-PRINTS

- [C4] **Rajesh R**, Rashen Fernando, and Ryan M. Corey, “Distance-Aware Source Separation via Multi-Microphone Spatial Features in Reverberant Rooms.”
- [C3] **Rajesh R**, Rashen Fernando, Padmanabhan Rajan, and Ryan M. Corey. “Bleed No More : Generative Interference Reduction for Musical Recordings.”
- [C2] Rashen Fernando, **Rajesh R**, Kanad Sarkar, Andrew Singer, and Ryan M. Corey. “Neural Particle on a Sphere to Estimate Image Source Locations for Room Impulse Response Interpolation.”
- [J2] **Rajesh R**, Rashen Fernando, and Ryan M. Corey. “End-to-End Learnable DWT Frontends for Multitrack Music Bleed Reduction.”
- [J1] **Rajesh R** and Padmanabhan Rajan. “GIRNet : A Graph-Based Interference Reduction in Live Microphone Recordings.”  

PROFESSIONAL EXPERIENCE

- Research Internship** Nov 2023 - Jan 2024
 Dubverse.ai, India
 Researcher : Audio Source Separation
- Data Science Engineer** Jan 2018 - May 2019
 DeTect Technologies Pvt Ltd, IITMRP, Chennai
 Developing an end-to-end report generation pipeline and creating PyQt-based software solutions for internal applications.
 - Tools : *PyQt5, Pandas, Matplotlib, OpenCV, Git*

TEACHING ASSISTANSHIPS

- EC317 Digital Signal Processing, UIC Spring 2025
- EE511 Computer Vision, IIT Mandi Fall 2023
- CS671 Deep Learning & Applications, IIT Mandi Spring 2023
- IC151 Computing and Data Science, IIT Mandi Fall 2022
- CS571 Programming Practicum (Signal Processing), IIT Mandi Fall 2022

ACADEMIC AWARDS	— University Rank - BE : Honoured with Medals, Shield, and a check worth of Rs. 10,000 — BE : Academic Excellence award in all eight semesters	
VOLUNTEERING & WORKSHOP	— 23rd International Conference on Music Information Retrieval (ISMIR 2022) held at Indian Institute of Science (IISc) Bengaluru - Volunteered. — CompMusic Workshop (A workshop to introduce the field of Computational Musicology while focusing on the study of Carnatic Music) Project : <i>Tonic Identification in Carnatic Music</i>  — 25th International Conference on Speech and Computer (SPECOM 2023) held at Hubli and Goa, India. — The Winter School on Speech and Audio Processing workshop in the theme of “Music for AI” took place in 2023 at the Indian Institute of Technology Kanpur.	
TECHNICAL SKILLS	Programming : Machine Learning : Audio & Signal Processing : Systems : UI : Version Control :	Python PyTorch, TensorFlow/Keras Librosa, Pyroomacoustics Linux, macOS, Windows PyQt Git
LANGUAGES	Tamil (Native), English (Fluent)	
REFERENCES AVAILABLE UPON REQUEST	Ryan M. Corey Padmanabhan Rajan	Assistant Professor, UIC  Associate Professor, IIT Mandi 